Isogenies on Oasis

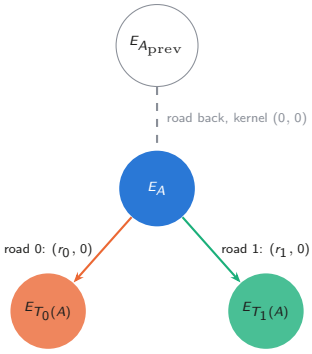
Qiskit Fall Fest 2026, Track 1

Three Qiskiteers

`github.com/atharva7-g/qff`

November 2026

The campus



- A spot is a curve $E_A: \; y^2 = x^3 + Ax^2 + x$, with $A \in \mathbb{F} = \mathbb{F}_p(i)$, $p \equiv 3 \pmod 4$.
- A road is a 2-isogeny. Its kernel is one 2-torsion point $(r, 0)$:

$$r_{0,1} = \frac{-A \pm \sqrt{A^2 - 4}}{2}, \quad \text{sorted by } \operatorname{key}(u + vi) = u + pv.$$

- Bit b takes the road through r_b:

$$T_b(A) = \operatorname{can}(2 - 4r_b^2), \quad \operatorname{can}(z) = \text{the smaller key of } \pm z.$$

- A password is a route $x = b_1 \cdots b_L$. It ends at

$$H(x) = j(E_{A_L}) = \frac{256(A_L^2 - 3)^3}{A_L^2 - 4}.$$

- Small campus: $p = 1019$, $A_0 = 977 + 297i$, $L = 12$, $j^* = 797 + 921i$.

Terms

The campus

- **Spot**: a curve E_A, a vertex of the graph.
- **Road**: a 2-isogeny, an edge. Each spot has 3 roads. A route never goes back along the road it arrived by, so each step has 2 choices.
- **Route**, **password**: the bits $x = b_1 \cdots b_L$; bit b_t picks the road at step t.
- **Maths department**: the start spot A_0.
- **ANC**: the target of Part I, every spot with $j(E_A) = j^*$.
- **Clock Tower**: the target of Part II, every spot with $j(E_A) \in \mathbb{F}_p$.

The method

- **Survey**: a classical map of the campus, built from p alone by walking each road once. Part I builds its circuit from it.
- **Query**: one use of the one-step circuit.
- $\{cx, rz, sx, x\}$: the IBM basis gate set we transpile to.
- **Djinn**, **volunteer** (Part IV): the spot names are hidden behind random labels. The Djinn returns the label at the end of a road; the volunteer reports whether a labelled spot is on the Clock Tower or at ANC.

The four parts

Part I **The password.** On the small campus ($p = 1019$, $L = 12$), find every route from A_0 to ANC with a quantum procedure, and give the confidence that the list is complete. Count the queries, and compare them with classical search and the best possible quantum search, in terms of $N = 2^L$ and the number of routes k.

Part II **The phone on the Clock Tower.** Give the quantum cost, as a power of p, of finding a route of length $\Theta(\log p)$ that ends on the Clock Tower. Prove the upper bound, naming the theorems used. Prove a matching lower bound in a stated model, and explain why none can be proved for the real arithmetic. Measure how often routes end there as p grows and fit the exponent. Run Grover on a campus where at most 1 in 32 routes succeeds, against theory and classical walking.

Part III **The Oasis map.** For $p = 1019$, build one step as a reversible circuit from arithmetic in $\mathbb{F}$ alone, with no table of roads: square root, key comparison, sign choice, and every ancilla cleaned. Check it on every spot and both bits. Give its cost at $p \approx 2^{256}$, with the Clock-Tower test and the Part II search.

Part IV **The Djinns.** The signboards are shuffled by a secret random permutation (the Djinn model). Say what still works, what stops working and why, which classical methods survive, and whether the Part II lower bound still holds.

Constraints

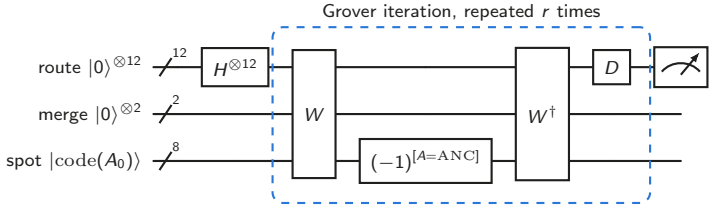
Search rules

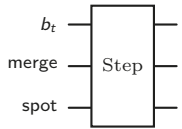
- The survey walks each road once (at most 2 moves per spot). It does not use A_0, L or j^*.
- After the survey: no path counting, no search backwards from ANC, no meet-in-the-middle, no route lists.
- The only classical work: walk one measured route once, to check it.
- The ANC test marks the codes of the ANC spots with a lookup on the spot register, inside the circuit. It does not evaluate H.
- If no route reaches ANC, the program reports it.

Build rules

- All circuits are built and run in Qiskit.
- Resources are given after transpiling to a stated basis: queries, qubits, two-qubit gates, depth. Part III also gives the Toffoli and T counts.
- One command reproduces every result. It takes p, A_0, L, j^*.
- One query is one use of the one-step circuit.

Part I: the circuit



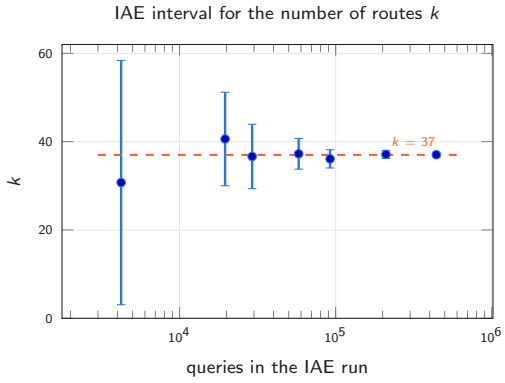


$W = \mathrm{Step}_{12} \cdots \mathrm{Step}_1$, step t reads bit b_t

- Survey: 255 spots, 510 roads. Each spot gets an 8-bit code.
- Two roads end at the same spot ($A = 6$), so the step alone is not one-to-one. Two merge qubits record which road arrived; the step is then a permutation of basis states.
- A single-target gate flips one bit, controlled by the other bits. Every permutation of n bits is a product of at most $2n - 1$ of them (De Vos and Van Rentergem, 2008). Our step uses 17.
- Oracle: W, phase -1 on the ANC codes, $W^\dagger$; 24 queries. $D = 2\,|s\rangle\langle s| - I$ is the Grover diffusion, $|s\rangle$ the uniform superposition.
- 22 qubits. In $\{cx, rz, sx, x\}$: one step 13,395 CX, depth 42,173.

L. K. Grover, *A fast quantum mechanical algorithm for database search*, STOC 1996.
A. De Vos, Y. Van Rentergem, *Young subgroups for reversible computers*, Adv. Math. Commun. 2 (2008).

Part I: counting the routes



- **IAE** (Iterative Amplitude Estimation): it estimates $a = k/N$ from circuits $Q^m A|0\rangle$ with Grover powers Q^m. Each round makes the confidence interval smaller (Appendix A1).
- We stop when the interval for $k = aN$ holds one integer. Result: $k = 37$ at confidence ≥ 0.99.
- $k = 0$ gives an interval at 0, and the program reports that no route reaches ANC.
- **Why IAE.** We first tried maximum-likelihood amplitude estimation. With few shots its likelihood had a second peak at $k \pm 1$ ($k = 5$ against 6 at $N = 256$). IAE keeps $(2m + 1)\theta_a$ inside one half-period, so this cannot happen. Counting by phase estimation needs a quantum Fourier transform and an extra register; IAE needs neither.

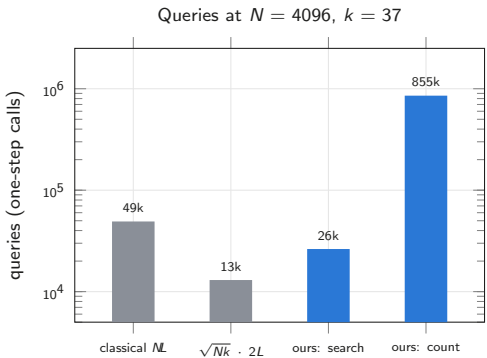
D. Grinko, J. Gacon, C. Zoufal, S. Woerner, *Iterative quantum amplitude estimation*, npj Quantum Inf. 7 (2021).
Y. Suzuki et al., *Amplitude estimation without phase estimation*, Quantum Inf. Process. 19 (2020).
G. Brassard, P. Høyer, A. Tapp, *Quantum counting*, ICALP 1998.

Part I: finding every route

- Grover with $r = 8$ iterations per run, 137 runs. Each measured route is checked once by one classical walk.
- A run succeeds with probability just under 1, so it can return a route that misses ANC. The check caught 1 such run, and it was discarded.
- All 37 routes are found. They end at 3 spots:

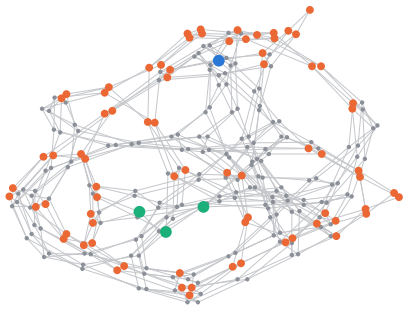
A_{12}	routes	j
$3 + 10i$	14	$797 + 921i$
$933 + 205i$	9	$797 + 921i$
$561 + 411i$	14	$797 + 921i$

- Chart: one-step calls. Grey: walk all N routes classically (NL), and the best possible quantum search for all k routes ($\sqrt{Nk} \cdot 2L$). Blue: our run.
- Our search is within $2\times$ of the best possible. The count costs the most: at $N = 4096$ the constants of IAE dominate (32 shots per round, the confidence terms). For large N the scaling, $\sqrt{Nk}$ against N, decides.



J. van Apeldoorn, S. Gribling, H. Nieuwboer, *Basic quantum subroutines: finding multiple marked elements and summing numbers*, Quantum 8 (2024).

Part II: the target



$p = 1019$: 255 spots, one connected graph. ● on the tower (75), ● ANC (3), ● A_0

- Clock Tower: $T_p = \{E : j(E) \in \mathbb{F}_p\}$.
- Deuring–Eichler theorem (counts the supersingular curves defined over $\mathbb{F}_p$): $|T_p| = 2h(-p)$ or $h(-p)$. Here $h(-p)$ is the class number of $\mathbb{Q}(\sqrt{-p})$.
- Siegel's theorem (growth of class numbers): $h(-p) = p^{1/2+o(1)}$. All spots: $|V_p| = p/12 + O(1)$.

$$\frac{|T_p|}{|V_p|} = p^{-1/2+o(1)}$$

- Small campus: 75 of 255 spots (29%). At $p \approx 2^{256}$: about 1 spot in 2^{125}.

M. Deuring, *Die Typen der Multiplikatorenringe elliptischer Funktionenkörper*, Abh. Math. Sem. Hamburg 14 (1941).
C. Delfs, S. D. Galbraith, *Computing isogenies between supersingular elliptic curves over $\mathbb{F}_p$*, Des. Codes Cryptogr. 78 (2016).
C. L. Siegel, *Über die Classenzahl quadratischer Zahlkörper*, Acta Arith. 1 (1935).

Part II: from spots to routes

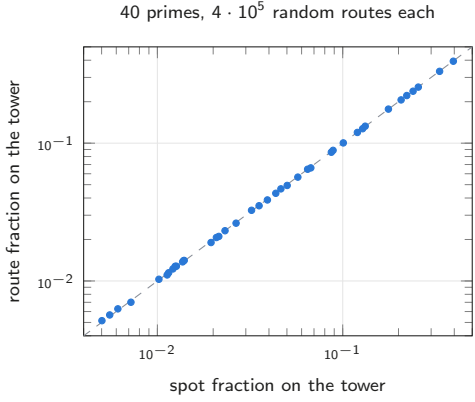
- The graph is connected (from any spot, the survey reaches all 255), 3-regular and Ramanujan: every nontrivial eigenvalue has $|\lambda| \le 2\sqrt{2}$, the best possible expansion (Pizer, 1990). A route is a walk on it without backtracking.
- Lubetzky–Peres (how fast such walks mix on Ramanujan graphs): the end of the walk is close to uniform in total variation distance,

$$d_{\mathrm{TV}} \le O\big(\sqrt{p}\, 2^{-L/2} L\big).$$

- With $L = \lceil 3 \log_2 p \rceil$ this is $\widetilde{O}(p^{-1})$, much less than the target fraction. So

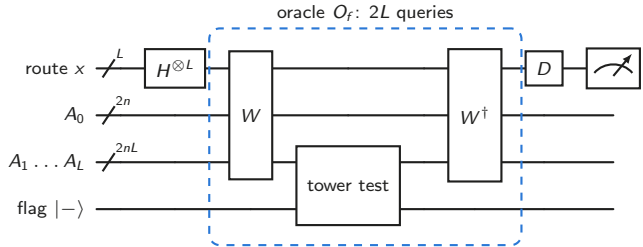
$$q_L = \Pr_x[f(x) = 1] = p^{-1/2+o(1)}.$$

- Data (right): route fraction = spot fraction, for 40 primes, already with $L = 2\lceil \log_2 p \rceil$.



A. K. Pizer, *Ramanujan graphs and Hecke operators*, Bull. AMS 23 (1990).
E. Lubetzky, Y. Peres, *Cutoff on all Ramanujan graphs*, GAFA 26 (2016).

Part II: the upper bound



- $O_f |x\rangle = (-1)^{f(x)} |x\rangle$, $f(x) = [j(E_{A_L(x)}) \in \mathbb{F}_p]$.
- With $\sin^2\theta = q_L$: after t iterations, $P(t) = \sin^2((2t+1)\theta)$.
- $t \approx \frac{\pi}{4\sqrt{q_L}} = p^{1/4+o(1)}$ iterations, each $2L = O(\log p)$ queries.
- q_L unknown: BBHT (Boyer–Brassard–Høyer–Tapp) runs Grover with random iteration counts; at most $\frac{9}{4}\sqrt{1/q_L}$ iterations on average.
- Classical random routes: $1/q_L = p^{1/2+o(1)}$.

M. Boyer, G. Brassard, P. Høyer, A. Tapp, *Tight bounds on quantum searching*, Fortschr. Phys. 46 (1998).

Part II: the lower bound

Black-box model

Input: an unknown $f : \{0,1\}^L \to \{0,1\}$ with k marked inputs. The only access is the oracle $O_f |x, z\rangle = |x, z \oplus f(x)\rangle$.

BBBV (Bennett–Bernstein–Brassard–Vazirani) and Zalka: every quantum algorithm with constant success needs

$$\Omega\left(\sqrt{N/k}\right) = \Omega\left(q_L^{-1/2}\right) = \Omega\left(p^{1/4-o(1)}\right)$$

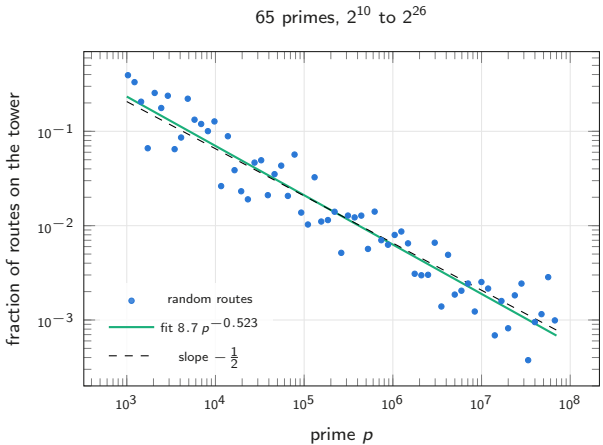
oracle queries. Grover meets this bound.

The real arithmetic

- The step is explicit field arithmetic. An algorithm can use this structure, so the black-box bound does not apply.
- Deciding whether a route to the tower exists is in NP (Appendix A3). BQP is the class of problems a quantum computer solves in polynomial time. A proof that no polynomial-time quantum algorithm exists would show NP $\not\subseteq$ BQP, and so P $\neq$ NP.
- No known algorithm beats $p^{1/4}$ under these rules (Appendix A4).

C. H. Bennett, E. Bernstein, G. Brassard, U. Vazirani, *Strengths and weaknesses of quantum computing*, SIAM J. Comput. 26 (1997).
C. Zalka, *Grover's quantum searching algorithm is optimal*, Phys. Rev. A 60 (1999).

Part II: data



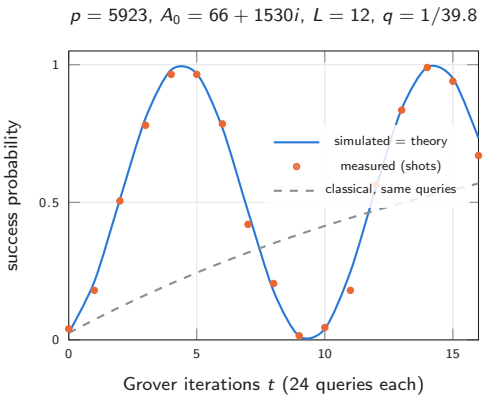
- Fit: $\widehat{q} \approx 8.7\,p^{-\alpha}$,

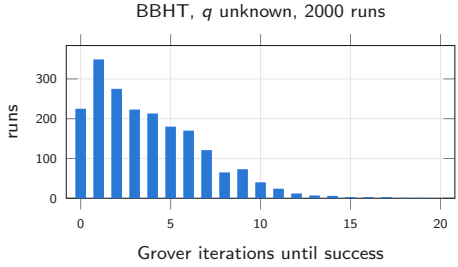
$$\widehat{\alpha} = 0.523 \pm 0.040$$

(65 primes, 2 standard errors).

- Exact spot fractions, 40 primes: $\widehat{\alpha} = 0.560 \pm 0.081$.
- The scatter comes from the class number $h(-p)$, not from noise: the same primes are high in both columns.
- Theory: $\alpha = \frac{1}{2}$.

Part II: demonstration

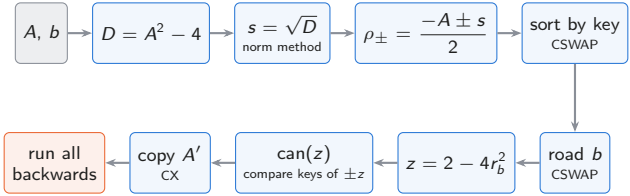




- Simulation = theory to 4 decimals. At $t = 4$: 0.981, against 0.205 classically.
- BBHT: on average 91 queries and 7.9 classical route checks. Classical random routes: 477 steps.
- 25 qubits. One iteration in $\{cx, rz, sx, x\}$: $3.40 \cdot 10^6$ CX, depth $1.11 \cdot 10^7$.

M. Boyer, G. Brassard, P. Høyer, A. Tapp, *Tight bounds on quantum searching*, Fortschr. Phys. 46 (1998).

Part III: one step from arithmetic



Square root from the norm

$D = \alpha + \beta i$. Two exponentiations in $\mathbb{F}_p$:

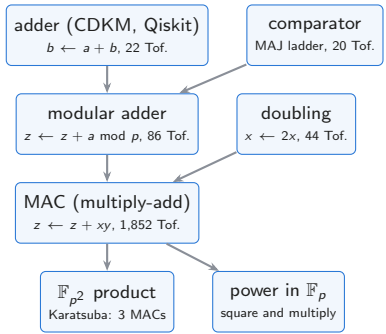
$$\lambda = (\alpha^2 + \beta^2)^{\frac{p+1}{4}}, \quad \delta = \tfrac{\alpha+\lambda}{2}, \quad r = \delta^{\frac{p-3}{4}}.$$

$y = r\delta$, $t = \beta r/2$, $q = [yr = 1]$: $s = y + ti$ if $q = 1$, else $s = -t + yi$. No inversion (Appendix A5).

- No branches: each condition is computed into a flag qubit, which controls the gates. CSWAP is a controlled swap.
- Any root s works: the key sort fixes the order of $\rho_\pm$.
- $|A, b\rangle|0\rangle \mapsto |A, b\rangle|T_b(A)\rangle$. Bennett's method returns all work qubits to 0: compute, copy A', run the computation backwards.
- Clock-Tower test without division: write $j = N/D$ and $\overline{a + bi} = a - bi$. Then $j \in \mathbb{F}_p \iff \overline{N}D = N\overline{D}$.

C. H. Bennett, *Logical reversibility of computation*, IBM J. Res. Dev. 17 (1973).

Part III: building blocks



Toffoli counts for $p = 1019$, $n = 10$ bits

- Gates: X, CX, Toffoli, CSWAP only.
- CDKM: the Cuccaro–Draper–Kutin–Moulton ripple-carry adder, built into Qiskit. The comparator keeps only its carry chain (MAJ gates).
- Modular adder: add, subtract p, add p back if the sign bit is 1, then clear the flag with the comparator.
- Doubling is a relabelling of qubits plus one subtraction of p.
- MAC: for each bit y_i, add $y_i \cdot x$, then double x. This gives n modular adders, so $O(n^2)$.
- Karatsuba: an $\mathbb{F}_{p^2}$ product from 3 products in $\mathbb{F}_p$ instead of 4.
- An exponentiation is $O(n)$ products, so one step is $O(n^3)$.

S. A. Cuccaro, T. G. Draper, S. A. Kutin, D. P. Moulton, *A new quantum ripple-carry addition circuit*, arXiv:quant-ph/0410184 (2004).
A. Karatsuba, Yu. Ofman, *Multiplication of multidigit numbers on automata*, Soviet Physics Doklady 7 (1963).

Part III: checks at $p = 1019$

Every input

- Qiskit Aer's matrix-product-state simulator, 587 qubits.
- Step: 510 inputs (255 spots, both bits). Output = classical $T_b(A)$, inputs unchanged, every work qubit 0. **510/510.**
- Clock-Tower test: **255/255** spots.
- 32 inputs on random spots at $p = 46351$: all correct.

Why basis states are enough

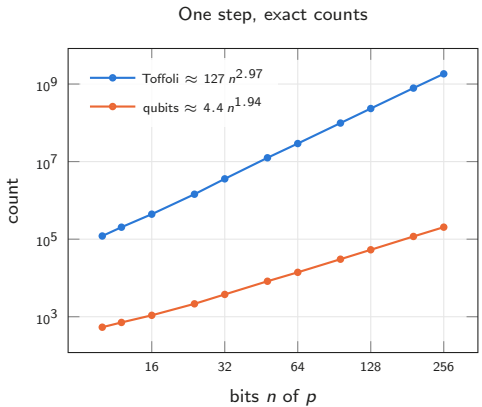
X, CX, Toffoli and CSWAP send basis states to basis states, so the circuit U is a permutation matrix. If $U|x\rangle|0\rangle = |x\rangle|f(x)\rangle$ for every x, then by linearity

$$U \sum_x c_x |x\rangle|0\rangle = \sum_x c_x |x\rangle|f(x)\rangle .$$

Phases

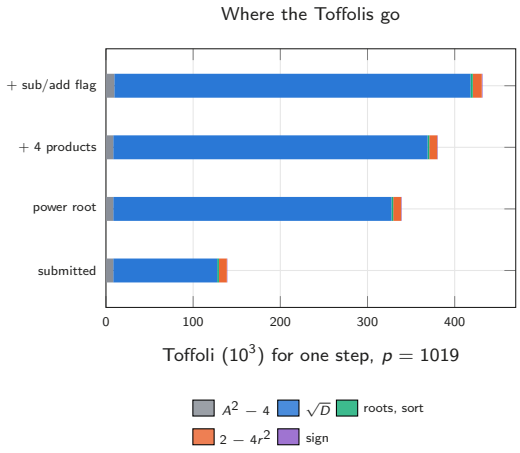
64 states $\cos w\,|x\rangle + e^{i\phi} \sin w\,|y\rangle$ go through the circuit and then the inverse of a reference circuit (Appendix A7). All return to $|0\rangle$. A deliberate phase error is detected.

Part III: cost at $p \approx 2^{256}$



one step, $n = 256$	
Toffoli	$1.83 \cdot 10^9$
T (7 per Toffoli)	$1.28 \cdot 10^{10}$
qubits	204,049
Part II search, $L = 2n = 512$	
target fraction q	$7/\sqrt{p} = 2^{-124.9}$
Grover iterations	$2^{62.1}$
Toffoli	$2^{102.9}$
two-qubit, $\{cx, rz, sx, x\}$	$2^{105.9}$
depth	$2^{104.4}$
logical qubits	466,000
classical random routes	$2^{133.9}$ steps

Part III: trade-offs, other algorithms



Each row changes one algorithm of the submitted circuit:

change	Toffoli	factor
submitted	139,236	1.00
root by powers in $\mathbb{F}_{p^2}$	339,162	2.44
+ $\mathbb{F}_{p^2}$ product as 4 products	380,506	2.73
+ flag by subtract and add back	431,914	3.10

Root by powers in $\mathbb{F}_{p^2}$: Algorithm 9 of Adj and Rodríguez-Henríquez (2014).

- The root is 86% of the step. In $\mathbb{F}_p$ a product is 1 MAC; in $\mathbb{F}_{p^2}$ it is 3 or 4.
- A comparison needs only the carry: $2n$ Toffolis, not $4(n+1)$.
- At $n = 256$: submitted $= 0.34\times$ the slowest version.

G. Adj, F. Rodríguez-Henríquez, *Square root computation over even extension fields*, IEEE Trans. Computers 63 (2014).

Part III: trade-offs, qubits and depth

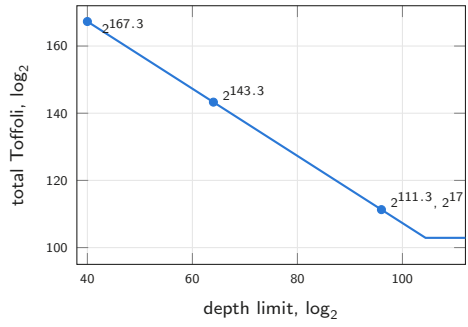
Qubits: which values to keep (pebbling)

Bennett's method keeps each intermediate value: about $1.5n$ registers of n qubits per exponentiation, so qubits $\sim n^2$.

strategy	registers	steps run
Bennett (ours)	T	$2T$
$\sqrt{T}$ checkpoints	$2\sqrt{T}$	$\approx 3T$
recursive halving	$\log_2 T$	$T^{1.58}$

We keep Bennett: it gives the fewest gates.

Part II search at $p \approx 2^{256}$ with a depth limit



With depth limit 2^m: $S = (2^{104.4-m})^2$ machines; total $\times\sqrt{S}$.

C. H. Bennett, *Logical reversibility of computation*, IBM J. Res. Dev. 17 (1973).
C. H. Bennett, *Time/space trade-offs for reversible computation*, SIAM J. Comput. 18 (1989).

Part IV: hidden names (Djinns)

The spot names are replaced by random labels. The Djinn returns the label at the end of a road; the volunteer reports whether a labelled spot is on the Clock Tower or at ANC.

What still works

- The graph is the same: degree, roads, cycles, expansion.
- Classical: random walks ($p^{1/2}$ tower tests), breadth- or depth-first search with a table of labels, cycle and collision search.
- Our Part I method: the survey asks the Djinn for every road ($O(N)$), then the step circuit is built from the survey.
- If the Djinn and volunteer accept superpositions (coherent access), Grover and quantum walks still give $p^{1/4}$.

What stops working

- Arithmetic on A: the Part III step circuit, the j test, the Frobenius map $a + bi \mapsto a - bi$.
- Methods from number theory (class groups, endomorphism rings).
- A Djinn or volunteer that answers classically takes one question at a time. No superposition, so no Grover.

Lower bound

The names are random, so the volunteer is close to a black box. The $\Omega(p^{1/4})$ bound needs one more condition: the tower set must not be visible from the graph shape.

F. Magniez, A. Nayak, J. Roland, M. Santha, *Search via quantum walk*, SIAM J. Comput. 40 (2011).

Summary

- **Part I:** all 37 passwords, k counted at confidence ≥ 0.99, survey rules kept.
- **Part II:** $p^{1/4+o(1)}$ quantum against $p^{1/2+o(1)}$ classical. Black-box lower bound; a lower bound for the real arithmetic would prove P $\neq$ NP. Fitted exponent 0.523 ± 0.040; Grover demo $=$ theory.
- **Part III:** one step from arithmetic only, 510/510 correct. At 2^{256}: $1.8 \cdot 10^9$ Toffolis per step, 2^{103} for the search.
- **Part IV:** the speed-up needs oracles that accept superpositions, not the spot names.

Reproduce

Code: `github.com/atharva7-g/qff`

```
uv run oasis --part 1 --p 1019 \
  --A0 "977 + 297i" --L 12 \
  --jstar "797 + 921i"
uv run oasis --part 2
uv run oasis --part 3
```

Results: `results/`

References (1/2)

- L. K. Grover, *A fast quantum mechanical algorithm for database search*, STOC 1996.
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- J. L. Delgado, *The Supersingular Isogeny Problem in Time and Memory $p^{1/3+o(1)}$, Unconditionally*, arXiv:2609.22018 (2026).

A1. Iterative Amplitude Estimation

- Problem: $A|0\rangle = \sqrt{a}\,|\text{good}\rangle + \sqrt{1-a}\,|\text{bad}\rangle$. Estimate $a = \sin^2\theta_a$.
- Grover operator $Q = AS_0A^\dagger S_{\text{good}}$, where S_0 and S_{good} flip the sign of $|0\rangle$ and of the good states. Then $Q^mA|0\rangle$ is good with probability $\sin^2((2m+1)\theta_a)$.
- Each round picks the largest power m that keeps $(2m+1)\theta_a$ inside one half-period of the current interval. It measures, and narrows the interval for θ_a (Clopper–Pearson: an exact confidence interval for a binomial proportion).
- Cost: $O\big(\frac{1}{\varepsilon}\log\frac{2}{\alpha}\log\frac{\pi}{4\varepsilon}\big)$ queries for error ε at confidence $1-\alpha$, with no QFT (Grinko, Gacon, Zoufal, Woerner 2021).
- Here: good = spot register at an ANC code, $a = k/N$. We halve ε until the interval for $k = aN$ holds one integer; $\alpha = 0.01$.

D. Grinko, J. Gacon, C. Zoufal, S. Woerner, *Iterative quantum amplitude estimation*, npj Quantum Inf. 7 (2021).
C. J. Clopper, E. S. Pearson, *The use of confidence or fiducial limits illustrated in the case of the binomial*, Biometrika 26 (1934).

A2. Theorems we use

Deuring–Eichler For $p \equiv 3 \pmod 4$, the number of supersingular $j \in \mathbb{F}_p$ is $2h(-p)$ if $p \equiv 3 \pmod 8$ and $h(-p)$ if $p \equiv 7 \pmod 8$.

Siegel $h(-p) = p^{1/2+o(1)}$.

Pizer The supersingular 2-isogeny graph is 3-regular and Ramanujan: every nontrivial eigenvalue has $|\lambda| \leq 2\sqrt{2}$.

Lubetzky–Peres For a walk without backtracking of length L on a 3-regular Ramanujan graph with N_E directed edges: $\left\|\tilde{\mu}_L/\pi_E - 1\right\|_{L^2}^2 \leq 2N_E 2^{-L}(8L^2+1)$.

BBBV, Zalka Search for one of k marked items among N with a black-box oracle needs $\Omega(\sqrt{N/k})$ queries; Grover is optimal.

BBHT With k unknown and $\lambda = 6/5$, the expected number of Grover iterations is at most $\frac{9}{4}\sqrt{N/k}$.

M. Deuring, *Die Typen der Multiplikatorenringe elliptischer Funktionenkörper*, Abh. Math. Sem. Hamburg 14 (1941).
C. Delfs, S. D. Galbraith, *Computing isogenies between supersingular elliptic curves over $\mathbb{F}_p$*, Des. Codes Cryptogr. 78 (2016).
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M. Boyer, G. Brassard, P. Høyer, A. Tapp, *Tight bounds on quantum searching*, Fortschr. Phys. 46 (1998).

A3. The Clock-Tower problem is in NP

Decision problem: given p and A_0, is there a route of length $L = O(\log p)$ that ends on the Clock Tower?

1. **Certificate:** the route $x = b_1 \cdots b_L$. Its length is $O(\log p)$ bits.
2. **Check:** L steps, each a fixed number of operations in $\mathbb{F}_{p^2}$ (one square root is one exponentiation), then compute j and test $\mathrm{Im}\, j = 0$. Each operation costs $\mathrm{poly}(\log p)$, so the check costs $\mathrm{poly}(\log p)$.
3. So the problem is in NP. The input size is $n = \log_2 p$.
4. Our cost $p^{1/4} = 2^{n/4}$ is exponential in n. A proof that no polynomial-time quantum algorithm exists would show NP $\not\subseteq$ BQP. Since P $\subseteq$ BQP, this gives P $\neq$ NP.
5. So an unconditional lower bound for the real arithmetic is out of reach today. Lower bounds exist only in a model with a black-box oracle.

A4. Known algorithms that use the arithmetic

algorithm	solves	cost	why not here
Biasse–Jao–Sankar (2014)	path between two curves, quantum	$\widetilde{O}(p^{1/4})$	same exponent as Grover
Page–Wesolowski (2023)	path, classical, from both ends	$\widetilde{O}(p^{1/2})$	meet-in-the-middle; forbidden by the rules
Delgado (2026)	one endomorphism, classical	$p^{1/3+o(1)}$	not a fixed-degree path; the path version needs GRH and gives longer paths
Wesolowski (2021)	path $\Leftrightarrow$ endomorphism ring	reduction	assumes GRH

None of these gives less than $p^{1/4}$ for a route of length $\Theta(\log p)$ from A_0 under the rules.

GRH: the generalized Riemann hypothesis. Endomorphism ring: the ring of isogenies from a curve to itself. Meet-in-the-middle: search from both ends and match.

J.-F. Biasse, D. Jao, A. Sankar, *A quantum algorithm for computing isogenies between supersingular elliptic curves*, INDOCRYPT 2014.
A. Page, B. Wesolowski, *The supersingular Endomorphism Ring and One Endomorphism problems are equivalent*, arXiv:2309.10432 (2023).
J. L. Delgado, *The Supersingular Isogeny Problem in Time and Memory $p^{1/3+o(1)}$, Unconditionally*, arXiv:2609.22018 (2026).
B. Wesolowski, *The supersingular isogeny path and endomorphism ring problems are equivalent*, arXiv:2111.01481 (2021).

A5. Proof of the square root from the norm

Let $s = x_0 + x_1 i$ with $s^2 = D = \alpha + \beta i$. Then $x_0^2 - x_1^2 = \alpha$, $2x_0x_1 = \beta$, and $(x_0^2 + x_1^2)^2 = \alpha^2 + \beta^2 = m$.

- m is a square in $\mathbb{F}_p$, so $\lambda = m^{(p+1)/4}$ gives $\lambda^2 = m$, and $x_0^2 + x_1^2 = \pm\lambda$.
- $\delta = (\alpha + \lambda)/2$ and $\delta' = (\alpha - \lambda)/2$ satisfy $\delta\delta' = -\beta^2/4$. Since -1 is not a square ($p \equiv 3 \bmod 4$), exactly one of them is a square (if $\beta \neq 0$).
- $r = \delta^{(p-3)/4}$, $y = r\delta$: then $y^2 = \delta^{(p+1)/2} = \pm\delta$ and $yr = \delta^{(p-1)/2} = \pm 1 = q$-flag.
- $q = 1$: $y^2 = \delta = x_0^2$, $x_1 = \beta/(2y) = \beta r/2 = t$, so $s = y + ti$.
- $q = 0$: $y^2 = -\delta = x_1^2$, $x_0 = \beta/(2y) = -\beta r/2 = -t$, so $s = -t + yi$.
- $\delta = 0$ only if $\beta = 0$ and α is not a square; then $\delta := \alpha$ gives $q = 0$, $s = yi$ with $y^2 = -\alpha$.

Checked on every nonzero square for 13 primes ≤ 83, and in the circuit on random squares at $p = 43$ and $p = 131$.

A6. Resources

Part I, $p = 1019$, $L = 12$, $\{cx, rz, sx, x\}$, level 1

	queries	CX	depth
one step	1	13,395	42,173
Grover iteration	24	319,106	1,009,282

22 qubits. Total: 854,784 queries to count, 26,304 to search.

Part II, $p = 5923$, $L = 12$, $\{cx, rz, sx, x\}$, level 1

	queries	CX	depth
one step	1	141,813	462,568
Grover iteration	24	3,401,466	11,100,684
search, $t = 4$	96	13,605,864	44,402,739

25 qubits.

Part III, $p = 1019$, $\{x, cx, ccx\}$; two-qubit in $\{cx, rz, sx, x\}$ = CX +6·Toffoli

	qubits	Toffoli	2-qubit	depth
step	587	139,236	1,127,164	307,933
tower test	192	37,794	–	63,148

T: 974,652 (7 per Toffoli) or 556,944 (4 per Toffoli).

Part III, $n = 256$

step: Toffoli / qubits	$1.83 \cdot 10^9$ / 204,049
tower test: Toffoli	$2.1 \cdot 10^7$
search: Toffoli / depth	$2^{102.9}$ / $2^{104.4}$

A7. Note on the phase test

- The test prepares $\cos w\,|0\rangle\,|x\rangle + e^{i\phi}\sin w\,|1\rangle\,|y\rangle$, runs the new circuit, then the inverse of the reference circuit, then undoes the preparation. The result is $|0\rangle$ with probability $1 - \sin^2(2w)\sin^2(\Delta\theta/2)$.
- A path of pairs through the inputs gives $\theta(x) = \theta(y)$ for all of them: the two circuits are equal up to one global phase.
- With equal amplitudes ($w = \pi/4$), Aer's matrix-product-state simulator lost the relative phase for some values of ϕ, also for a circuit followed by its own inverse. Equal amplitudes give two equal Schmidt coefficients (the singular values across a cut of the state), so the simulator's singular value decomposition has no unique answer.
- With $w = 0.55$ every case passes. A deliberate error (Z on b) is found on every pair with different b, and only there.